BIO 280 PATHOPHYSIOLOGY · JOYCE UNIVERSITY

Dr. Garg's Exam 4 Master Study Guide

Modules 7 & 8 — Genetics & the Endocrine System

ⓘ What Exam 4 covers

This unit is about **the body's blueprint and its chemical messengers**. **Module 7 (Genetics)** covers DNA and inheritance — alleles, genotype vs phenotype, the three inheritance patterns, Punnett squares, chromosomal disorders (Down, Klinefelter, Turner), nondisjunction, and teratogens. **Module 8 (Endocrine)** covers hormones and their disorders — feedback loops, levels of dysfunction, growth hormone, ADH (DI vs SIADH), cortisol (Cushing's vs Addison's), thyroid/parathyroid, and diabetes mellitus. Exam format: **50 questions (multiple choice + select-all-that-apply)**, proctored with Respondus LockDown Browser.

ⓘ How this guide works

- **Click any topic card** in the navigator to jump to that section. Click a title bar to expand or close it.
- **All answers are hidden** — try first, then click "Show answer." Practice includes **multiple choice** and **select-all-that-apply (SATA)**, like the real exam.
- **Each topic has:** clear explanation, a mnemonic, a "common confusion" warning, practice questions, a clinical scenario, and a self-check.
- **Color code:** **purple = Module 7 (Genetics)**, **teal = Module 8 (Endocrine)**.
- **To save as PDF:** use the button above, or press *Ctrl+P* / *Cmd+P* → "Save as PDF." All sections expand for printing.

□Suggested 7-Day Study Schedule ▼

Day	Focus
Day 1	Topics 1–2 (genetics foundations & vocabulary; cell division & nondisjunction).
Day 2	Topic 3 (chromosomal disorders: Down, Klinefelter, Turner) + karyotypes.
Day 3	Topics 4–5 (inheritance patterns; Punnett square practice — do every worked example).
Day 4	Topic 6 (teratogens/TORCH) + review all of Module 7.
Day 5	Topics 7–9 (feedback loops & levels; growth hormone; ADH — DI vs SIADH).
Day 6	Topics 10–12 (cortisol/adrenal; thyroid/parathyroid; diabetes mellitus).
Day 7	Cheat sheet + comparison tables. Self-test with hidden answers.

□Topic Navigator

TOPIC 1 · MODULE 7
Genetics Foundations & Vocabulary

TOPIC 2 · MODULE 7
Cell Division & Nondisjunction

TOPIC 3 · MODULE 7
Chromosomal Disorders & Karyotypes

TOPIC 4 · MODULE 7
Inheritance Patterns

TOPIC 5 · MODULE 7
Punnett Squares: Step by Step

TOPIC 6 · MODULE 7
Teratogens & TORCH

TOPIC 7 · MODULE 8

TOPIC 8 · MODULE 8

TOPIC 9 · MODULE 8

Endocrine Foundations & Feedback Loops

Growth Hormone Disorders

ADH Disorders: DI vs SIADH

TOPIC 10 · MODULE 8

Cortisol & Adrenal: Cushing's vs Addison's

TOPIC 11 · MODULE 8

Thyroid & Parathyroid Disorders

TOPIC 12 · MODULE 8

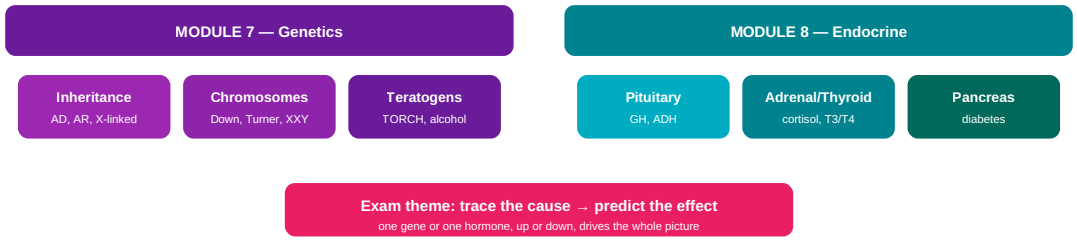
Diabetes Mellitus: Type 1, 2, DKA & HHS

□Exam Cheat Sheet

□Don't-Mix-These-Up

□Glossary

□The Big Picture: Two Modules



TOPIC 1 □Genetics Foundations & Vocabulary MUST KNOW ▼

The vocabulary you must own

Genetics questions live and die on definitions. Lock these in first — every Punnett-square and inheritance question reuses them.

Term	Definition
Gene	A stretch of DNA that codes for one protein — the unit of heredity.
Allele	A version (variant) of a gene at a given spot. You inherit two — one from each parent.
Locus	The gene's "address" (parking spot) on the chromosome.
Genotype	The actual allele code (e.g., BB, Bb, bb) — the "recipe."
Phenotype	The physical, observable trait (e.g., brown eyes) — the "cake."
Homozygous	Two identical alleles — BB (dominant) or bb (recessive).
Heterozygous	Two different alleles — Bb. The dominant allele is the one expressed.
Karyotype	A photographed map of a person's chromosomes, arranged by size/shape to spot numerical or structural defects.

Humans have **46 chromosomes** = **23 pairs** — 22 pairs of **autosomes** + 1 pair of **sex chromosomes** (XX female, XY male). **Homologous chromosomes** are the matching pair (one from each parent) carrying the same genes at the same loci.

Single-gene vs polygenic traits

- **Single-gene trait:** one gene, two possible outcomes — you have it or you don't (e.g., cystic fibrosis, sickle cell).
- **Polygenic (multifactorial) trait: many genes + environment** → a spectrum of outcomes. Examples: height, cardiovascular health, and later-life diseases like **diabetes, hypertension, coronary artery disease, cancer**, and some mental illness. Early-life multifactorial examples: cleft lip/palate, clubfoot, congenital hip dislocation, congenital heart disease.

Mutations

A **mutation** is a change in the DNA sequence that can be passed on. Two main causes: an **extra/missing nucleotide** (shifts the whole sequence) or a **substitution** (one base swapped). Results range from **no change** at all to a **new gene variant** — which may be harmless, beneficial (sickle trait resists malaria), or harmful (cancer).

> **Memory hook**
> **Genotype = the recipe; Phenotype = the cake.** "Homo" = same (BB/bb), "Hetero" = different (Bb). **Poly**genic = **poly** = many genes + environment.

> **⚠ Common confusion**
> Genotype vs phenotype: **Bb and BB look the same** (both brown eyes) but have different genotypes. A heterozygous person can be a silent **carrier** of a recessive disease while showing a normal phenotype.

Practice Questions

1. A person has the genotype Bb, where B (brown) is dominant over b (blue). What is their eye-color phenotype, and could they pass on the blue allele?

   Show answer

   > Phenotype = **brown eyes** (the dominant B is expressed). Yes — they are heterozygous, so they carry the recessive b and can pass it to offspring.

2. Which trait is polygenic rather than single-gene: cystic fibrosis or adult height?

   Show answer

   > **Adult height** — it's controlled by many genes plus environment (nutrition). Cystic fibrosis is a single-gene trait.

3. **SATA:** Which are true of a karyotype? (a) shows all 46 chromosomes, (b) detects numerical defects like trisomy, (c) detects structural defects like translocations, (d) sequences individual genes.

   Show answer

   > **a, b, c.** A karyotype is a picture of the chromosomes (number, size, shape) — it spots extra/missing or rearranged chromosomes. It does **not** sequence individual genes (d).

Clinical scenario

A genetic counselor tells a couple that they are both unaffected "carriers" for the same condition. **What does carrier mean, and which kind of inheritance is implied?**

*A carrier is **heterozygous** (one normal + one mutated allele) with a **normal phenotype** who can still pass the mutation on. Carriers are the hallmark of **autosomal recessive** inheritance — disease only appears when a child inherits two recessive alleles.*

TOPIC 2

Cell Division & Nondisjunction

HIGH YIELD

Mitosis vs meiosis — know the outcomes

	Mitosis	Meiosis
Purpose	Growth & repair (replace worn-out cells)	Make sperm & eggs (reproduction)
Result	2 identical daughter cells	4 cells, each genetically unique
Chromosomes	**Diploid** → **diploid** (2n → 2n)	**Diploid** → **haploid** (2n → n)
Variation	None (clones)	Crossing over → genetic diversity

Why the halving matters: a **haploid** egg (n) + a **haploid** sperm (n) combine at fertilization to restore a **diploid** (2n) zygote. If gametes weren't halved, the chromosome count would double every generation.

Nondisjunction — the key error

Nondisjunction = "failure to separate." During meiosis, chromosomes are supposed to split evenly. When they don't, one gamete gets an **extra** chromosome and another gets **none**. After fertilization with a normal gamete, the embryo ends up with:

- **Trisomy** = one extra chromosome (47 total) → e.g., Trisomy 21 (Down), XXY (Klinefelter).
- **Monosomy** = one missing chromosome (45 total) → e.g., Monosomy X / 45,XO (Turner).

Risk of nondisjunction **increases with maternal age** (older eggs). It is the most common cause of **numerical** chromosomal disorders. (Contrast: **structural** abnormalities = deletions, duplications, inversions, translocations.)

> **Memory hook**
> **Mitosis = identical (clones for repair). Meiosis = makes eggs/sperm & halves.**
> **Tri**somy = **three** (extra, 47). **Mono**somy = **one** (missing, 45).

⚠ **Common confusion**

Gene disorder vs chromosomal disorder: a gene disorder = a mutation in **one gene** (cystic fibrosis); a chromosomal disorder = a change in the **number or structure of whole chromosomes** (Down syndrome). Nondisjunction causes the chromosomal kind.

Practice Questions

1. A sperm cell formed with two copies of chromosome 21 fertilizes a normal egg. How many chromosome 21 copies does the zygote have, and what is the condition?

   Show answer

   Three copies (47 chromosomes total) = Trisomy 21 = Down syndrome. The extra came from the nondisjunction error in the sperm.

2. Which type of cell division produces genetic variation through crossing over — mitosis or meiosis?

   Show answer

   Meiosis. Crossing over exchanges DNA segments, which is why siblings differ. Mitosis makes identical clones.

3. Nondisjunction during meiosis most directly causes which category of disorder, and name its two numerical outcomes.

   Show answer

   Numerical chromosomal disorders — outcomes are **trisomy** (extra chromosome) and **monosomy** (missing chromosome).

Clinical scenario

A 42-year-old is told her pregnancy has a higher risk of a chromosomal disorder than it did at 25. **Why does maternal age raise this risk?**

*Older oocytes are more prone to **nondisjunction** during meiosis, so the chance of producing a gamete with an extra or missing chromosome (leading to trisomy or monosomy) rises with maternal age.*

TOPIC 3

Chromosomal Disorders & Karyotypes

MUST KNOW

All three classic chromosomal disorders come from **nondisjunction**. The exam wants the karyotype, the sex, and the hallmark features of each.

Disorder	Karyotype	Sex	Hallmark features
Down syndrome (Trisomy 21)	47, +21	Either	Flattened face/nose, upslanting eyes, **single simian (palmar) crease**, growth + mild–moderate intellectual delay. ↑ risk of **heart defects, leukemia, diabetes**.
Klinefelter (Trisomy XXY)	47, XXY	**Male**	Tall, normal intelligence, **low testosterone** → small testes, **infertility**, delayed/absent puberty, gynecomastia, sparse body hair, broad hips. Most common genetic cause of **male infertility**.
Turner (Monosomy XO)	45, XO	**Female**	**Short stature, webbed neck**, low-set ears, shield chest, **heart defects (coarctation)**, low thyroid, streak (immature) ovaries → **infertility**. Normal intelligence with some spatial/math difficulty.

The mechanism in one line

Down = extra autosome 21 (an extra non-sex chromosome). **Klinefelter = extra X in a male** (XXY → too little testosterone, feminizing features). **Turner = missing X in a female** (XO → underdeveloped ovaries, short stature, heart/thyroid issues).

Memory hook

Turner = "Tiny" female (short, one X, 45). **Klinefelter = "Klassic" tall male** with an extra X (47, XXY). **Simian crease + flat face** → Down (Trisomy 21).

⚠ **Common confusion**

Klinefelter (47,XXY) is the **extra-X male**; Turner (45,XO) is the **missing-X female**. Both cause infertility, but one is a trisomy and one is a monosomy — opposite directions.

Practice Questions

1. A karyotype shows 47 chromosomes with three copies of chromosome 21. Name the condition and one common associated health risk.

   Show answer

   Trisomy 21 (Down syndrome). Associated risks include congenital **heart defects**, **leukemia**, and diabetes.

2. A tall adolescent male with small testes, gynecomastia, and infertility most likely has which chromosomal disorder and karyotype?

   Show answer

   Klinefelter syndrome, 47,XXY. The extra X lowers testosterone, producing feminizing features and infertility.

3. **SATA:** Which are features of Turner syndrome? (a) 45,XO karyotype, (b) male, (c) short stature, (d) webbed neck, (e) coarctation of the aorta.

   Show answer

   a, c, d, e. Turner is a **female** (45,XO) with short stature, webbed neck, and heart defects (coarctation). (b) is false — there is no Y chromosome.

Clinical scenario

A newborn girl is short for length, has extra neck skin folds, and a murmur from aortic narrowing. Karyotype returns 45,XO. **What condition is this, and what reproductive/endocrine issues should the team anticipate?**

Turner syndrome. *Anticipate* ***streak (immature) ovaries*** *→ low estrogen, absent puberty and* ***infertility****, plus* ***low thyroid hormone****. Management often includes growth hormone and later estrogen replacement, with cardiac/thyroid monitoring.*

TOPIC 4

□Inheritance Patterns (AD, AR, X-linked)

MUST KNOW ▼

The single biggest genetics topic. For each pattern, know: **how many copies are needed**, **which sexes are affected**, the **pedigree clue**, and the **common conditions**.

Pattern	Rule	Pedigree clue	Common conditions
Autosomal Dominant (AD)	**1 dominant allele** expresses it (BB or Bb affected; only bb unaffected)	Appears in **every generation** (vertical); affects M & F equally	**Huntington's, Marfan**, neurofibromatosis, familial hypercholesterolemia, polycystic kidney disease
Autosomal Recessive (AR)	**2 recessive alleles** needed (bb affected; Bb = silent **carrier**)	Often **skips generations**; two unaffected carriers → affected child; M & F equally	**C**ystic fibrosis, **P**KU, **S**ickle cell, **T**ay-Sachs
X-linked Recessive	Gene on X. **Males (XY) need only 1** bad X to be affected; females need 2	**Mostly males affected**; carrier mothers pass to sons; affected dad → all daughters carriers, no sons	**H**emophilia A/B, red-green **C**olor blindness, **D**uchenne muscular dystrophy

The key probabilities

- **AD, one affected (Bb) × unaffected (bb):** each child **50%** affected.
- **AD, both heterozygous (Bb × Bb): 75% affected** (1 BB + 2 Bb), 25% unaffected.
- **AR, two carriers (Bb × Bb): 25% affected**, **50% carriers**, 25% unaffected.
- **X-linked, carrier mom × unaffected dad: 50% of sons** affected; **50% of daughters** carriers.
- **X-linked, affected dad × unaffected mom: 0% of sons** affected (dad gives sons his Y), **100% of daughters** carriers.

Spotlight conditions

- **Marfan (AD):** connective-tissue disorder → tall, long fingers (arachnodactyly), pectus deformity, lens dislocation; **aortic dilation/dissection is the deadly feature**.

- **Huntington's (AD):** progressive brain-cell degeneration in mid-adulthood → **chorea** (dance-like movements), cognitive decline, psychiatric symptoms; 50% inheritance risk.
- **Cystic fibrosis (AR):** CFTR defect → thick mucus in lungs, pancreas, reproductive tract.
- **Duchenne MD (X-linked):** dystrophin loss → progressive muscle wasting in young boys.

Memory hooks

AR mnemonic: **"Can't Pass Sick Traits"** → **C**ystic fibrosis, **P**KU, **S**ickle cell, **T**ay-Sachs. X-linked: **"He Can't see/Duchenne"** → **H**emophilia, **C**olor blindness, **D**uchenne — mostly boys. AD shows up **every generation** (Dominant = Dynasty).

⚠ Common confusion (classic exam trick)

A male cannot be a "carrier" of an X-linked recessive trait. He has only one X — if it carries the gene he is **affected**, not a carrier. Only females (XX) can be silent carriers.

Practice Questions

1. A disease appears in every generation and affects sons and daughters equally; one affected parent tends to have about half affected children. Which inheritance pattern?

   Show answer

   Autosomal dominant. Vertical (every-generation) transmission, both sexes, ~50% of offspring affected from one heterozygous parent.

2. Two unaffected parents have a child with cystic fibrosis. What were the parents' genotypes, and what is the chance the next child is affected?

   Show answer

   Both parents are **carriers (Bb)** for this autosomal recessive disease. Each pregnancy: **25% affected**, 50% carrier, 25% unaffected.

3. A condition affects mostly boys; an affected man's sons are all healthy but his daughters are all carriers. Which pattern, and name one example?

Show answer

X-linked recessive. Fathers give sons their Y (so sons are spared) and their X to every daughter (making them carriers). Examples: hemophilia, Duchenne MD, color blindness.

4. **SATA:** Which conditions are autosomal recessive? (a) Huntington's, (b) cystic fibrosis, (c) sickle cell disease, (d) Marfan, (e) Tay-Sachs.

Show answer

b, c, e. Cystic fibrosis, sickle cell, and Tay-Sachs are AR. Huntington's and Marfan are autosomal **dominant**.

Clinical scenario

A 40-year-old with a strong family history (a parent and grandparent affected) develops mood changes and involuntary jerky "dance-like" movements. **What condition and inheritance pattern, and what is the risk to each of his children?**

Huntington's disease — autosomal dominant. *Hallmark is* ***chorea*** *with cognitive/psychiatric decline. Because one affected parent carries one dominant allele,* ***each child has a 50% chance*** *of inheriting it — which is why genetic counseling is central.*

TOPIC 5

□Punnett Squares: Step by Step

HIGH YIELD ▼

Punnett-square questions are guaranteed points if you follow a fixed routine. Don't guess — build the grid. (Below, X^a = normal X and X^h = disease-carrying X.)

□The 4-step method

1. **Identify the pattern** from key words (every generation = dominant; skips/carriers = recessive; mostly males = X-linked).
2. **Write each parent's genotype** (uppercase = dominant, lowercase = recessive; for X-linked, attach alleles to X and remember Y carries nothing).
3. **Fill the grid** — one allele from each parent into each box.
4. **Count boxes** → convert to a ratio/percentage and answer exactly what's asked (affected? carrier? unaffected? by sex?).

Worked example 1 — Autosomal recessive (carrier × carrier)

Both parents are carriers (Aa × Aa). "A" is the normal dominant allele; "a" is recessive (disease). The affected box is shaded.

Aa × Aa	A	a
A	**AA**	**Aa**
a	**Aa**	**aa**

Results: **1 AA (unaffected), 2 Aa (carriers), 1 aa (affected) → 25% affected, 50% carriers, 25% unaffected**.

Worked example 2 — Autosomal dominant (affected × unaffected)

One parent has an AD condition and is heterozygous (Aa); the other is unaffected (aa). "A" causes the disease (shaded = affected).

Aa × aa	A	a
a	**Aa**	**aa**
a	**Aa**	**aa**

Results: **2 Aa (affected), 2 aa (unaffected) → 50% of children affected**. (If both parents were Aa × Aa, it would be 75% affected.)

Worked example 3 — X-linked recessive (affected father × unaffected mother)

Father has hemophilia (X^hY); mother is an unaffected non-carrier (X^aX^a). Shaded = carrier daughters.

$X^hY \times X^aX^a$	X^h	Y
X^a	X^aX^h	X^aY
X^a	X^aX^h	X^aY

All **daughters are carriers** (X^aX^h); all **sons are unaffected** (X^aY). An **affected father has 0% affected sons** but 100% carrier daughters — he passes his Y (not his X) to sons.

> ⚠ **Common confusion**
> Read the **exact question**. "Chance of being *affected*" vs "*carrier*" vs "*unaffected*" are different counts on the same grid. If it asks "of the **sons**," only count the male boxes.

Practice Questions

1. Two carriers for an autosomal recessive disease (Aa × Aa) have a child. What is the probability the child is an unaffected non-carrier (AA)?

   Show answer

   > **25%** — only the single AA box out of four is a fully unaffected non-carrier.

2. Two people heterozygous for Marfan syndrome (autosomal dominant, Aa × Aa) have a child. What is the chance the child is **unaffected**?

   Show answer

   > **25%** — only the aa box is unaffected; 75% (1 AA + 2 Aa) are affected.

3. A carrier mother (X^aX^h) and an unaffected father (X^aY) have a son. What is the chance that son has the X-linked disease?

Show answer

50% of sons are affected — each son has a 1-in-2 chance of inheriting the mother's X^h.

Clinical scenario

Both partners are sickle-cell carriers and ask, "What are the odds our baby has sickle cell disease?" **Walk through the answer.**

*Sickle cell is autosomal recessive. Carrier × carrier (Aa × Aa) → **25% affected (aa), 50% carriers (Aa), 25% unaffected (AA)**. Each pregnancy is independent — the 25% doesn't "use up."*

TOPIC 6

□□Teratogens & TORCH

HIGH YIELD ▼

A **teratogen** is any substance or agent that causes **fetal malformation**. The fetus is most vulnerable during **organogenesis — roughly days 15–60** after conception (weeks 3–8), when organs are forming.

- **Before day 15:** "all-or-nothing" — either no effect or miscarriage.
- **Days 15–60:** highest risk of **structural birth defects**.
- **After day 60:** more likely to affect **growth or brain development** than structure.

Common teratogens to recognize

Category	Examples & effects
Alcohol	Fetal alcohol syndrome (facial changes, growth restriction, lifelong neurobehavioral issues). **No safe amount.**
Medications	Thalidomide (limb defects), isotretinoin/Accutane (brain/heart/face), valproic acid (neural tube defects).
Recreational drugs	Cocaine/meth/opioids → placental abruption, preterm birth, withdrawal, growth restriction.
Tobacco	Vasoconstriction → low birth weight, preterm birth, ↑ SIDS risk.
Chemicals	**Mercury** (contaminated fish), lead, pesticides → brain/nervous-system damage.
Infections (TORCH)	See mnemonic below → microcephaly, deafness, blindness, heart defects.
Radiation	Ionizing radiation (X-rays, radiotherapy) → growth restriction, microcephaly, skeletal defects.

□Memory hook — TORCH infections

Toxoplasmosis · **O**ther (syphilis, varicella, HIV, parvovirus) · **R**ubella · **C**ytomegalovirus (CMV) · **H**erpes simplex. Window reminder: organs form weeks 3–8, so that's the structural-defect danger zone.

⚠ Common confusion

Timing changes the damage: an exposure during **weeks 3–8** tends to cause

structural malformations; the same exposure later mostly affects **growth and the brain**, not organ structure.

Practice Questions

1. During which window is the fetus at highest risk for structural birth defects, and why?

   Show answer

   About days 15–60 (weeks 3–8) — the period of **organogenesis**, when organ systems are forming and most vulnerable.

2. What does the TORCH acronym stand for?

   Show answer

   Toxoplasmosis, **O**ther (syphilis, varicella, HIV, parvovirus), **R**ubella, **C**ytomegalovirus, **H**erpes simplex.

3. **SATA:** Which are recognized teratogens? (a) alcohol, (b) folic acid, (c) isotretinoin (Accutane), (d) mercury, (e) rubella.

   Show answer

   a, c, d, e. Alcohol, isotretinoin, mercury, and rubella are teratogens. **Folic acid (b) is protective** — it helps prevent neural tube defects.

Clinical scenario

At a first prenatal visit, a patient reports occasional wine, takes isotretinoin for acne, and eats swordfish weekly. **Which exposures are teratogenic and what counseling is needed?**

*All three carry risk: **alcohol** (no safe amount → fetal alcohol syndrome), **isotretinoin** (severe craniofacial/cardiac/CNS defects — stop immediately), and **high-mercury fish** (neurotoxic). Counsel to stop alcohol and isotretinoin and limit high-mercury fish.*

TOPIC 7

⚖ Endocrine Foundations & Feedback Loops

MUST KNOW

The chain of command

Most hormones run through the **hypothalamus → pituitary → target gland** axis. The hypothalamus is the "control center"; the **anterior pituitary** makes tropic hormones; the **posterior pituitary** stores ADH & oxytocin (made by the hypothalamus).

- **Anterior pituitary hormones:** GH, prolactin, **ACTH** (→ cortisol), **TSH** (→ thyroid), FSH, LH.
- **Posterior pituitary (stores): ADH** (water retention) and oxytocin.

Feedback loops — negative vs positive

	Negative feedback (the default)	Positive feedback (rare)
Action	The product **shuts off** its own production — keeps levels in a set range (like a thermostat).	The product **amplifies** the response until an endpoint is reached.
Example	Rising cortisol turns off ACTH/CRH; rising T3/T4 turns off TSH.	Oxytocin in labor (contractions → more oxytocin → stronger contractions); the LH surge.

Levels of dysfunction — where is the problem?

Level	Where	Example
Primary	The **target gland** itself (adrenal, thyroid)	Addison's (adrenal destroyed); Hashimoto's (thyroid destroyed)
Secondary	The **pituitary** (too much/little tropic hormone)	Pituitary adenoma; pituitary failure to make ACTH/TSH
Tertiary	The **hypothalamus** (releasing-hormone problem)	Hypothalamic CRH/TRH deficiency

The ACTH–cortisol logic (high-yield)

Trace it with feedback. In a **primary** adrenal problem, ACTH moves **opposite** to cortisol; in a **secondary** (pituitary) problem, ACTH and cortisol move **together**.

- **Primary high cortisol** (adrenal adenoma): cortisol ↑ → negative feedback → **ACTH & CRH low**.
- **Secondary high cortisol** (pituitary makes too much ACTH): **ACTH high → cortisol high** (they rise together).
- **Primary low cortisol** (Addison's): cortisol ↓ → no negative feedback → **ACTH & CRH high** (this drives the bronze skin).
- **Secondary low cortisol** (pituitary can't make ACTH): **ACTH low → cortisol low** (both low).

> **Memory hook**
> **Negative = "Nip it"** (product stops itself). **Positive = "Push it"** (product amplifies; think oxytocin in labor). For levels: **1° gland, 2° pituitary, 3° hypothalamus** — the higher the number, the higher up the chain.

> **⚠ Common confusion**
> When you see a "what happens to ACTH?" question, first decide **primary vs secondary**. Primary gland problems make the pituitary hormone go the **opposite** way (feedback); pituitary problems make both go the **same** way.

Practice Questions

1. An adrenal adenoma autonomously secretes high cortisol. What happens to ACTH and CRH, and why?

   Show answer

   > **ACTH and CRH both fall (low).** High cortisol triggers **negative feedback** on the pituitary and hypothalamus. This is a **primary** problem (the gland itself).

2. Which is an example of positive feedback?

   Show answer

   > **Oxytocin during labor** (or the LH surge) — the response amplifies itself until an endpoint (delivery/ovulation). Most endocrine control is negative feedback.

3. A pituitary tumor fails to make ACTH, causing low cortisol. Is this primary, secondary, or tertiary dysfunction, and what is the ACTH level?

   Show answer

   Secondary (pituitary). **ACTH is low**, so cortisol is low — both move together because the problem is upstream at the pituitary.

Clinical scenario

Labs show **low cortisol with HIGH ACTH**. **Where is the lesion, and what classic skin finding follows?**

*The gland is the problem — this is **primary adrenal insufficiency (Addison's)**. Low cortisol removes negative feedback, so ACTH climbs. High ACTH stimulates melanocytes → **bronze hyperpigmentation** of the skin.*

TOPIC 8

Growth Hormone Disorders

HIGH YIELD

Growth hormone (GH / somatotropin) from the anterior pituitary drives cell growth, protein synthesis, lipid mobilization, and growth of bone, cartilage, and muscle. Disorders are about **too little** or **too much** — and **timing relative to the growth plates** is everything.

Disorder	GH	Timing	Key features
Dwarfism (GH deficiency)	↓ Low	Childhood	**Short stature with proportional features**, delayed puberty; causes idiopathic, congenital, or acquired.
Gigantism	↑ High	**Before** growth plates close (children)	Excess GH (usually a pituitary adenoma) → **very tall**, long bones grow lengthwise.
Acromegaly	↑ High	**After** growth plates close (adults)	Bones thicken instead of lengthen → **large hands/feet, coarse facial features, enlarged jaw/nose, deepening voice**; rings/shoes stop fitting.

The deciding factor: both gigantism and acromegaly are **GH excess** (usually a pituitary adenoma). If the excess starts **before** the growth plates fuse → **gigantism** (tall). If it starts **after** they fuse → **acromegaly** (broad/thick, not taller).

Memory hook

Giants grow tall while young; Acromegaly is an Adult's Already-closed plates → they grow **out**, not up (big jaw, hands, feet).

⚠ Common confusion

Gigantism and acromegaly are the **same cause** (GH excess) — only the **age/growth-plate status** differs. Don't pick "different hormones."

Practice Questions

1. An adult develops enlarging hands, a protruding jaw, and shoes that no longer fit. Which GH disorder, and why not increased height?

Show answer

> **Acromegaly** — GH excess **after** the growth plates have fused, so bones thicken/widen rather than lengthen. No new height is gained.

2. A child has GH excess from a pituitary adenoma before puberty. What is the expected result?

   Show answer

   > **Gigantism** — with open growth plates, long bones grow lengthwise → very tall stature.

3. A child with low GH shows short stature but normally proportioned body features. What is this, and name a possible cause.

   Show answer

   > **Growth hormone deficiency (dwarfism)** — proportional short stature. Causes can be idiopathic, congenital, or acquired (e.g., pituitary damage).

Clinical scenario

A 45-year-old reports gradually coarsening facial features, sweating, and headaches; an MRI shows a pituitary mass. **What is the diagnosis and the underlying hormone problem?**

Acromegaly *from a* ***GH-secreting pituitary adenoma****. In an adult (closed plates) the excess GH thickens bone and soft tissue — coarse features, enlarged hands/feet/jaw — rather than increasing height.*

TOPIC 9

□ADH Disorders: DI vs SIADH

MUST KNOW ▼

ADH (antidiuretic hormone / vasopressin) tells the kidneys to **retain water**. The two disorders are mirror images — one has too little ADH, the other too much.

	Diabetes Insipidus (DI)	SIADH
ADH	**Too little** ADH	**Too much** ADH
Water	Kidneys **dump water**	Kidneys **retain water**
Urine	**Huge volumes of dilute** urine (polyuria)	**Scant, concentrated** urine
Signs	Extreme **thirst**, dehydration, **hypernatremia** (concentrated blood), hypotension	Fluid overload, **hyponatremia** (dilute blood), weight gain, confusion/ headache
Blood Na^+	**High** (water lost, sodium concentrated)	**Low** (water retained, sodium diluted)

How DI resembles diabetes mellitus

DI and diabetes mellitus share the name "diabetes" (meaning "to pass through") because both cause **polyuria and polydipsia** (lots of urine + lots of thirst). The difference: in **DM** the urine is loaded with **glucose**; in **DI** the urine is **dilute and glucose-free** — the problem is water handling, not sugar.

□Memory hook

DI = "Drains Inside out" (no ADH → pee out floods of dilute urine → Na^+ goes **up**). **SIADH = "Soaked, Sodium-low"** (too much ADH → retain water → Na^+ goes **down**). They sit on opposite ends of the sodium seesaw.

⚠ Common confusion

Anchor on **sodium**: DI → **hyper**natremia (losing water concentrates the blood). SIADH → **hypo**natremia (retaining water dilutes it). If you get the water direction, the sodium and symptoms follow.

Practice Questions

1. A patient produces large volumes of very dilute urine, is intensely thirsty, and has a high serum sodium. Too much or too little ADH, and which disorder?

   Show answer

   Too little ADH = Diabetes Insipidus. Kidneys can't retain water, so dilute urine pours out, the blood concentrates (hypernatremia), and thirst surges.

2. A post-op patient becomes confused with low urine output, weight gain, and serum sodium of 122 mEq/L. Which ADH disorder?

   Show answer

   SIADH — excess ADH retains water, diluting sodium (hyponatremia) and causing fluid overload and neuro changes.

3. How is diabetes insipidus similar to diabetes mellitus, and how do their urines differ?

   Show answer

   Both cause **polyuria + polydipsia**. In DM the urine is **full of glucose**; in DI it is **dilute and glucose-free** (a water-handling problem, not a sugar problem).

Clinical scenario

After a head injury, a patient suddenly makes 8 liters of pale urine a day and is severely thirsty; sodium is 152 mEq/L. **What is going on and why?**

Central diabetes insipidus *— pituitary/hypothalamic injury cut ADH output, so the kidneys can't concentrate urine. Massive water loss →* ***hypernatremia*** *and intense thirst. Treat with desmopressin (synthetic ADH) and fluids.*

TOPIC 10

Cortisol & Adrenal: Cushing's vs Addison's

MUST KNOW

The **adrenal cortex** (stimulated by **ACTH**) makes **cortisol** (glucocorticoid), **aldosterone** (mineralocorticoid), and androgens. Cortisol raises blood glucose, dampens inflammation/immunity, and helps maintain blood pressure. The two big disorders are opposites: **too much (Cushing's)** vs **too little (Addison's)**.

	Cushing's (HYPER)	Addison's (HYPO)
Problem	**Too much** cortisol (± aldosterone)	**Too little** cortisol & aldosterone
Cause	Chronic steroid use (most common), pituitary or adrenal adenoma	**Autoimmune destruction** of the adrenal gland
Signs	**Weight gain** (moon face, buffalo hump, central obesity), **hyperglycemia**, thin skin/easy bruising, purple striae, hypertension, ↑ **infection risk**, muscle wasting	**Hypotension**, fatigue, weight loss, salt craving, **hyperkalemia**, hyponatremia, **bronze skin**, hypoglycemia

Why Addison's causes bronze skin

In **primary** Addison's, the adrenal can't make cortisol, so **negative feedback is lost** and the pituitary pumps out **high ACTH**. ACTH is made from the same precursor as **melanocyte-stimulating hormone**, so high ACTH → **increased skin pigmentation (bronzing)**.

Addisonian crisis — the emergency

A sudden, life-threatening cortisol shortage → severe **hypotension/shock**, hypoglycemia, hyperkalemia. Triggers: **abruptly stopping steroids** (never stop without tapering), major stress/infection/surgery in someone with adrenal insufficiency. **Why tapering matters:** chronic oral steroids suppress the body's own production; stopping suddenly leaves a gap with no cortisol → crisis.

Pheochromocytoma (adrenal medulla)

A tumor of the **adrenal medulla** that oversecretes **catecholamines** (epinephrine/norepinephrine). Classic triad: **episodic severe headache, palpitations, and sweating (diaphoresis)** with **dangerously high, spiking blood pressure**.

Memory hooks

Cushing's = "Cushion" (extra padding: moon face, buffalo hump, central fat). **Addison's = "Add-a-stone"** looks the opposite — thin, wasted, low BP, bronze. Pheochromocytoma triad: the **3 P's — Pain (headache), Palpitations, Perspiration**.

⚠ Common confusion

Cortisol disorders flip glucose and potassium in opposite directions: Cushing's → **high glucose, low potassium**; Addison's → **low glucose, high potassium**. And never stop steroids abruptly — that precipitates an **Addisonian crisis**.

Practice Questions

1. A patient on long-term prednisone develops a round "moon" face, central obesity, high blood sugar, and easy bruising. Which disorder?

   Show answer

   Cushing's syndrome — excess cortisol (here from exogenous steroids). Hallmarks: moon face, buffalo hump, central obesity, hyperglycemia, thin skin/bruising.

2. Why do patients with primary Addison's disease develop bronze-colored skin?

   Show answer

   Low cortisol removes negative feedback → **ACTH rises**. ACTH shares a precursor with melanocyte-stimulating hormone, so high ACTH **increases skin pigmentation**.

3. **SATA:** Which can trigger an Addisonian crisis? (a) abruptly stopping chronic steroids, (b) major infection/surgery in adrenal insufficiency, (c) gradual steroid taper, (d) severe physiologic stress.

   Show answer

   a, b, d. Sudden steroid withdrawal and major stress/illness can precipitate crisis. A **gradual taper (c) prevents** it by letting natural cortisol production recover.

4. A patient has episodic pounding headaches, palpitations, sweating, and BP spiking to 220/120. Which adrenal tumor?

Show answer

Pheochromocytoma — an adrenal medulla tumor releasing catecholamines; the 3 P's (pain/headache, palpitations, perspiration) with severe paroxysmal hypertension.

Clinical scenario

A patient with known adrenal insufficiency stops their hydrocortisone for 3 days, then collapses with BP 76/40, glucose 48, and potassium 6.1. **What is happening and why?**

Addisonian crisis *from abrupt steroid withdrawal. With no cortisol/aldosterone, the patient develops* ***profound hypotension/shock, hypoglycemia, and hyperkalemia****. It's an emergency — needs immediate IV steroids and fluids. Lesson:* ***taper, never stop abruptly.***

TOPIC 11

□Thyroid & Parathyroid Disorders

MUST KNOW ▼

Thyroid hormone (T3/T4) sets the body's metabolic "speed." **Hypo** = everything **slows down**; **hyper** = everything **speeds up**. The pituitary controls it with **TSH**.

	Hypothyroidism (slow)	Hyperthyroidism (fast)
Cause	**Hashimoto's** (autoimmune), iodine deficiency, congenital	**Graves'** (autoimmune); less often nodules, thyroiditis, iodine excess
Signs	**Fatigue, weight gain, cold intolerance**, bradycardia, constipation, dry skin/hair, depression, slowed thinking	**Weight loss, heat intolerance**, tachycardia/palpitations, anxiety, tremor, diarrhea, exophthalmos (Graves')
Primary labs	**↑ TSH, ↓ T3/T4**	**↓ TSH, ↑ T3/T4**

Reading the labs (primary thyroid disease)

In a **primary** thyroid problem, **TSH moves opposite to T3/T4** (negative feedback). **Low** thyroid hormone → the pituitary screams for more → **high TSH** (hypothyroid). **High** thyroid hormone → the pituitary goes quiet → **low TSH** (hyperthyroid).

Goiter

A **goiter** is an enlarged, swollen thyroid — and it can occur in **either** hypo- or hyperthyroidism (classically iodine deficiency and Graves'). Enlargement alone doesn't tell you the hormone level — labs are needed.

Hypoparathyroidism — the calcium gland

The **parathyroid glands** raise blood calcium (via PTH). **Hypoparathyroidism → low PTH → HYPOcalcemia.** Low calcium causes **neuromuscular irritability**: tingling, muscle cramps, **tetany**, and positive **Trousseau's sign** (carpal spasm when a BP cuff is inflated) and Chvostek's sign. A classic cause is **accidental damage/removal during thyroid surgery** (the parathyroids sit on the thyroid).

□Memory hooks

Hypo = "hibernate" (cold, tired, slow, gaining weight). **Hyper = "hyperactive"** (hot,

jittery, fast, losing weight). Primary thyroid labs: **TSH and T3/T4 point opposite ways**. **Hypo**parathyroid → **hypo**calcemia → **cramps/tetany**.

⚠ **Common confusion**

Students flip the TSH. In **primary hypo**thyroidism TSH is **HIGH** (gland is failing, pituitary pushes harder). A goiter can appear in **both** hypo and hyper — you still need T3/T4 + TSH to tell them apart.

Practice Questions

1. A patient is fatigued, cold, constipated, gaining weight, and has a heart rate of 54 with a goiter. T4 is low. What will the TSH be in primary disease?

   Show answer

   High (above normal). Primary hypothyroidism: low T3/T4 removes feedback, so the pituitary raises **TSH** trying to stimulate the failing gland.

2. **SATA:** Which are signs of hyperthyroidism? (a) weight loss, (b) heat intolerance, (c) bradycardia, (d) tremor/anxiety, (e) diarrhea.

   Show answer

   a, b, d, e. Hyperthyroidism speeds everything up. (c) is wrong — expect **tachycardia**, not bradycardia.

3. After a thyroidectomy, a patient develops tingling lips and a carpal spasm when the BP cuff inflates. Which gland was likely injured and what electrolyte is low?

   Show answer

   The **parathyroid glands** were damaged → **hypoparathyroidism** → **hypocalcemia**. Low calcium causes neuromuscular irritability (positive Trousseau's sign, tetany).

Clinical scenario

A young woman has palpitations, a 12-lb weight loss despite eating more, heat intolerance, bulging eyes, and a diffuse goiter. **What is the diagnosis and expected TSH/T4 pattern?**

Graves' disease (hyperthyroidism). *Autoantibodies overstimulate the thyroid →* ***high T3/T4*** *and* ***suppressed (low) TSH****. Exophthalmos and a diffuse goiter are classic Graves' findings.*

TOPIC 12

Diabetes Mellitus: Type 1, 2, DKA & HHS

MUST KNOW

Insulin (from pancreatic **beta cells**) lets glucose **into** cells and lowers blood sugar. Without working insulin, glucose piles up in the blood while cells "starve."

What insulin normally does

- Moves **glucose into cells** (muscle, fat, liver) for energy.
- Promotes **glycogen, fat, and protein storage** (anabolic, "fed-state" hormone).
- **Lowers blood glucose** — the opposite of glucagon, cortisol, and epinephrine.

	Type 1 DM	Type 2 DM
Patho	**Autoimmune** destruction of beta cells → **no insulin**	Cells become **insulin-resistant** (± relative insulin lack)
Onset / body	Usually younger, often thin; abrupt	Usually adult, often overweight; gradual
Risk factors	Genetic/autoimmune predisposition	**Obesity, inactivity, family history, age, metabolic syndrome**
Emergency	**DKA**	**HHS**

The classic signs — the "polys"

Polyuria (osmotic diuresis from glucose spilling into urine) → dehydration → **polydipsia** (thirst); **polyphagia** (hunger because cells can't use glucose); plus **weight loss** (body burns fat/muscle), **blurry vision** (lens swelling), and fatigue.

DKA vs HHS — the two crises

	DKA (Diabetic Ketoacidosis)	HHS (Hyperosmolar Hyperglycemic State)
Usually	Type 1	Type 2
Glucose	High — usually **> 250 mg/dL**	Very high — usually **> 600 mg/dL**
Ketones/ acidosis	**Yes** — ketones in urine, **metabolic acidosis**, **Kussmaul**	**No/minimal** ketones (enough insulin to block ketosis)

	(deep, rapid) breathing, fruity breath	
Hallmark	Acidosis + ketosis	**Profound dehydration, "thickened" (hyperosmolar) blood**, altered mental status

Complications & gestational diabetes

- **Chronic complications: macrovascular** (heart attack, stroke, peripheral artery disease) and **microvascular** (retinopathy → blindness, nephropathy → kidney failure, neuropathy → numbness/ulcers), plus poor wound healing and infections.
- **Gestational diabetes** = insulin resistance in pregnancy (the placenta blocks insulin; mom can't keep up in late pregnancy). Complications: a **large baby (macrosomia)**, **polyhydramnios**, birth injury, neonatal hypoglycemia, and higher future Type 2 risk for mother and child.

Memory hooks

The **3 P's**: **P**olyuria, **P**olydipsia, **P**olyphagia. **Type 1 = no insulin** (immune attack, thin, DKA). **Type 2 = resistant insulin** (overweight, HHS). **DKA has Ketones + aciDosis + Kussmaul**; **HHS is Hugely high glucose, Hyperosmolar, no ketones**.

⚠ Common confusion

Both DKA and HHS have high glucose — the separator is **ketones + acidosis (DKA)** vs **none + extreme dehydration with the highest glucose (HHS)**. And DM shares polyuria/polydipsia with **DI**, but DM's urine is full of **glucose** (Topic 9).

Practice Questions

1. A thin teenager presents with rapid deep breathing, fruity breath, glucose 480, ketones in the urine, and a low blood pH. Which type of diabetes and which crisis?

   Show answer

   Type 1 DM in DKA. No insulin → the body burns fat → ketones → metabolic acidosis with compensatory **Kussmaul respirations** and fruity breath.

2. An older adult with Type 2 diabetes has glucose 820, severe dehydration, confusion, and **no** significant ketones. Which emergency?

Show answer

HHS. There's enough insulin to prevent ketosis, but glucose climbs extremely high → profound osmotic dehydration, hyperosmolar blood, and altered mental status.

3. List three functions of insulin.

   Show answer

   Moves **glucose into cells**, promotes **storage** of glycogen/fat/protein (anabolic), and **lowers blood glucose**.

4. **SATA:** Which are recognized risk factors for Type 2 diabetes? (a) obesity, (b) physical inactivity, (c) autoimmune beta-cell destruction, (d) family history, (e) older age.

   Show answer

   a, b, d, e. Obesity, inactivity, family history, and age drive Type 2. **Autoimmune beta-cell destruction (c) is Type 1.**

Clinical scenario

At 28 weeks, a pregnant patient screens positive for gestational diabetes. **Why does pregnancy cause this, and what fetal complications follow?**

*Placental hormones cause **insulin resistance**; in late pregnancy the mother can't make enough insulin to keep up, so glucose rises. Extra glucose crosses to the fetus → **macrosomia (large baby)**, **polyhydramnios**, birth injury risk, and **neonatal hypoglycemia** after delivery.*

Exam Cheat Sheet — Night-Before Review

Module 7 — Genetics

- **Genotype** = allele code (BB/Bb/bb); **phenotype** = what you see. **Homo**zygous = same; **hetero**zygous = different (Bb = carrier).
- **Mitosis** = identical clones (2n→2n, repair). **Meiosis** = gametes, halved (2n→n, variation).

- **Nondisjunction** (meiosis error) → **trisomy** (extra, 47) or **monosomy** (missing, 45). Risk ↑ with maternal age.
- **Down** = Trisomy 21 (flat face, simian crease, heart defects/leukemia). **Klinefelter** = 47,XXY (tall infertile male, gynecomastia). **Turner** = 45,XO (short female, webbed neck, coarctation, infertile).
- **Karyotype** = picture of chromosomes; finds numerical + structural defects.
- **AD** = 1 allele, every generation (Huntington's-chorea, Marfan-aorta). **AR** = 2 alleles, carriers, "Can't Pass Sick Traits" (CF, PKU, Sickle cell, Tay-Sachs). **X-linked** = mostly males (Hemophilia, Color blindness, Duchenne).
- **Punnett %:** AD Aa×aa = 50% affected; AD/AR Aa×Aa = 75%/25% (dominant) or 25% affected + 50% carrier (recessive); affected dad (X-linked) → 0% sons, 100% carrier daughters.
- **Teratogens:** worst days 15–60 (organogenesis). **TORCH** = Toxo, Other, Rubella, CMV, Herpes. Alcohol = no safe amount. Folic acid is protective, not a teratogen.
- **Polygenic** = many genes + environment (height, diabetes, HTN, CAD, cancer).

Module 8 — Endocrine

- **Negative feedback** = product shuts itself off (default). **Positive** = amplifies (oxytocin/labor, LH surge).
- **Levels:** 1° = gland, 2° = pituitary, 3° = hypothalamus. **Primary** gland problem → pituitary hormone goes **opposite**; **secondary** → both go **same** way.
- **ACTH/cortisol:** primary high cortisol → ACTH low; Addison's (low cortisol) → ACTH high → **bronze skin**.
- **GH:** deficiency = dwarfism (proportional short). Excess **before** plates close = **gigantism**; **after** = **acromegaly** (big hands/jaw/feet).
- **ADH: DI** = too little ADH → floods of dilute urine, **high Na^+**. **SIADH** = too much ADH → retain water, **low Na^+**.
- **Cushing's** (high cortisol) = moon face, central obesity, **high glucose, low K^+**, thin skin. **Addison's** (low cortisol) = hypotension, weight loss, **low glucose, high K^+**, bronze skin. Never stop steroids abruptly → **Addisonian crisis**.
- **Pheochromocytoma** (adrenal medulla) = 3 P's: Pain (headache), Palpitations, Perspiration + spiking BP.
- **Thyroid (primary):** hypo → **↑TSH, ↓T3/T4** (cold, slow, gaining; Hashimoto's). Hyper → **↓TSH, ↑T3/T4** (hot, fast, losing; Graves'). **Goiter** = enlarged thyroid in either.
- **Hypoparathyroid** → **hypocalcemia** → tetany/Trousseau's (often post-thyroidectomy).
- **Diabetes:** 3 P's (polyuria, polydipsia, polyphagia). T1 = autoimmune, no insulin, **DKA** (ketones, acidosis, Kussmaul, glucose >250). T2 = insulin resistance, **HHS**

(glucose >600, no ketones, profound dehydration). Gestational → macrosomia, polyhydramnios.

□Don't-Mix-These-Up

Easily confused	The key difference
Klinefelter vs Turner	Klinefelter = **extra X, male** (47,XXY). Turner = **missing X, female** (45,XO).
Trisomy vs Monosomy	Trisomy = **extra** chromosome (47). Monosomy = **missing** (45).
Dominant vs Recessive	Dominant = **1 copy**, every generation. Recessive = **2 copies**, carriers, can skip.
Gigantism vs Acromegaly	Same cause (GH excess) — **before** plates close = tall (gigantism); **after** = thick (acromegaly).
DI vs SIADH	DI = too little ADH, dilute urine, **high Na^+**. SIADH = too much ADH, retained water, **low Na^+**.
Cushing's vs Addison's	Cushing's = too much cortisol (cushioned, high glucose/low K). Addison's = too little (wasted, bronze, low glucose/high K).
Hypo vs Hyperthyroid	Hypo = slow/cold/gain, ↑**TSH**. Hyper = fast/hot/lose, ↓**TSH** (primary disease).
DKA vs HHS	DKA (T1) = ketones + acidosis + Kussmaul, glucose >250. HHS (T2) = no ketones, glucose >600, profound dehydration.
DI vs Diabetes Mellitus	Both: polyuria + polydipsia. DM urine = **full of glucose**; DI urine = **dilute, no glucose**.

□Glossary

Allele — a version of a gene; you inherit two, one per parent.

Genotype / Phenotype — the allele code vs the observable trait.

Homozygous / Heterozygous — two identical alleles vs two different (heterozygous recessive = carrier).

Karyotype — a photographed map of all 46 chromosomes used to find numerical/ structural defects.

Nondisjunction — failure of chromosomes to separate in meiosis → trisomy or monosomy.

Trisomy / Monosomy — one extra (47) vs one missing (45) chromosome.

Polygenic (multifactorial) — a trait from many genes plus environment (height, diabetes).

Carrier — a heterozygous person with a normal phenotype who can pass on a recessive allele.

Teratogen — an agent that causes fetal malformation (worst during days 15–60).

ACTH — anterior pituitary hormone that stimulates the adrenal cortex to make cortisol.

ADH (vasopressin) — hormone that makes the kidneys retain water.

Negative feedback — a product inhibits its own production to hold levels steady.

Cortisol — adrenal glucocorticoid; raises glucose, suppresses inflammation/immunity, supports BP.

Goiter — an enlarged thyroid gland (can occur in hypo- or hyperthyroidism).

DKA / HHS — diabetic ketoacidosis (ketones + acidosis, often T1) vs hyperosmolar hyperglycemic state (extreme glucose, dehydration, often T2).

Kussmaul respirations — deep, rapid breathing that blows off CO_2 to compensate for metabolic acidosis in DKA.

Pheochromocytoma — adrenal-medulla tumor releasing catecholamines → headache, palpitations, sweating, hypertension.

Final Study Tips

- **Genetics is rules + practice.** Learn the pattern rules cold, then drill Punnett squares until the percentages (100% / 75% / 50% / 25%) are automatic.
- **Endocrine is cause → effect.** For every hormone, ask "too much or too little?" and the signs follow. Most disorders are just one hormone too high (hyper) or too low (hypo).

- **Lock in the opposite pairs:** Cushing's vs Addison's, hyper- vs hypothyroidism, DI vs SIADH, DKA vs HHS, trisomy vs monosomy. The "don't-mix-these-up" table is built for this.
- **Know the ACTH–cortisol logic:** in a *primary* adrenal problem the feedback runs opposite to the cortisol; in a *secondary* (pituitary) problem ACTH and cortisol move together.
- **Memorize the lettered sets:** TORCH, the autosomal-recessive mnemonic, and the diabetes "polys" (polyuria, polydipsia, polyphagia).
- **Practice SATA carefully** — read each option as its own true/false statement; partial credit usually isn't given.
- **Use the cheat sheet** the night before AND the morning of the exam, then self-test with the answers hidden. Format: 50 questions, Respondus LockDown Browser — prep your space ahead of time.

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Click the **Save as PDF** button at the top, or press **Ctrl+P** (Windows) / **Cmd+P** (Mac) and choose "Save as PDF." All collapsed sections expand automatically for printing.

You've got this. Master the inheritance rules and the "too much vs too little" logic, and the scenarios practically answer themselves.

Dr. Garg · BIO 280 Pathophysiology · Joyce University